Game Theory
Lecture 1:
Decision making under uncertainty and expected utility theorem[1]

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Outline

1. Administrative things
2. What is GT?
3. Decision making under uncertainty
4. Review and Exercises

Me

I am

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- interests: microeconomic theory
  contract theory, mechanism design, game theory, industrial organization, health economics

You?

Course I

- course website:
  `https://schottmueller.github.io/gt/gt2016.html`

Course II

- workload: 7.5 ECTS $\approx$ 200 h work
  1. attending lectures (14*3=42 h)
  2. work through your notes/textbook/paper after each lecture (14*3=42 h)
     - minimum: go through the slides and your notes until you can answer all "review questions"
     - standard: minimum + read the indicated readings
     - premium: occassionally there will be * readings
  3. work through exercises after each lecture (14*2=28 h)
     - minimum: try for 1 hour to solve the exercises
     - standard: try for 2 hours to solve the exercises
     - premium: occassionally there will be * exercises, do more exercises from the books
  4. midterm exam ($\approx$ 20 h)
     - 2 weeks, take home, groups of 2-3 people
  5. revision/exam preparation at the end of the course ($\approx$ 60 h)
     - oral exam: you draw one topic; explain topic + an exercise
  6. just for fun (very optional): some python scripts on website

Course III

- what you should know:
  - game theory:
    some intuitive understanding of
    - games
    - strategies
    - dominance
    - Nash equilibrium

    is quite helpful
  - maths:
    - solving simple equations (linear, quadratic)
    - taking derivatives and (unconstrained) maximization
    - functions: continuity, concavity/convexity
    - sequences and convergence of sequences (see handout online)
    - properties of sets (convex, closed/open) and basic set operations (union, intersection)
    - basic probabilities and expected value

Course IV

- related other courses:
  many other courses use game theory tools
  - mechanism design
  - advanced industrial organization
  - contract theory and the economics of organizations
  - corporate finance theory
  - ...
- lecture schedule: `https://schottmueller.github.io/gt/lectureplan2016.html`

Reading

- main textbooks (both available online for free!)
  - Osborne and Rubinstein "A course in Game Theory"
  - Maschler, Solan and Zamir "Game Theory"
- alternative sources
  - Fudenberg and Tirole "Game Theory"
  - Mas-Colell, Whinston, Greene "Microeconomic Theory"
- 2 handouts (notation, maths)
  - read before next lecture
- (parts of) a few papers (links on course website)

What is game theory? I

- game theory is a set of tools to help us *understand* the phenomena that we observe when decision makers interact ("comprehension")
- tools: various models and solution concepts
- real world —(1)— model —(2)— real world
  (1): modelling, simplification, representation; (judgement)
  (2): interpretation; (judgement)
  our main focus: model (solution, properties of the solution etc.)

Goal and method

- better understand real life situations of interactive decision making
- method: model as game and analyze how rational players would play

What is game theory? II

- implicit idea: insights and results help to understand more complicated settings (real world)

homo rationalis

- rational players, reasoning strategically:
  - rational:
    - well defined objective (function)
    - sole motivation is to achieve objectives (maximize objective function)
    - unlimited analytical ability
  - strategic:
    - knowing that other players are similarly rational
    - take this into account when reasoning

Decision making under uncertainty I

Motivation:

	C	D
C	2,2	0,3
D	3,0	1,1

Table: prisoner's dilemma

- What do the numbers in the game table actually mean?
- What if the other player plays C and D with 50% probability? How to evaluate that?
- can we model a rational decision maker as expected utility maximizer?

Decision making under uncertainty II

- today: no game, just decision problem of 1 decision maker under uncertainty
- basic setup: a decision maker has to choose among lotteries over outcomes in a set C
  - set of *outcomes* $C = \{c_1, c_2 \dots c_n\}$
  - a *simple lottery* L is a probability distribution $(p_1, p_2 \dots p_n)$ with $p_i \geq 0$ and $\sum_{i=1}^{n} p_i = 1$ where p_i is the probability of outcome c_i

Example (vacation lottery)

You book a vacation in the south. Depending on the weather your vacation has the outcomes
$C = \{\text{lying on the beach}, \text{ stuck in the hotel room}\}$. Given the weather forecast you assign probabilities $(0.9, 0.1)$ to the two possible outcomes.

Decision making under uncertainty III

- we start from "preferences" (rationality seems to imply that you know what you want)
  - preferences are a representation of (hypothetical) choice (which is the true primitive)
  - preferences do not denote "happiness" or anything like this (unless you assume that agents always choose what makes them "most happy")
- the decision maker has a *complete and transitive* preference relation $\succeq$ on the set of all simple lotteries

Decision making under uncertainty IV

Example (vacation lottery 2)

- third outcome: "being stuck at home", i.e. $C = \{$lying on the beach, stuck in hotel room, stuck at home$\}$
- probabiltiy 0.2 that your tour operator goes bankrupt before you go on holidays (and 0.8 that your holiday goes through)
- compound lottery: with probability $\alpha_1 = 0.8$ you get the vacation lottery; with probability 0.2 you get the "lottery" that puts all probability on the outcome "stuck at home"

A *compound lottery* $(L_1, \ldots, L_K; \alpha_1, \ldots, \alpha_K)$ yields with probability α_k the simple lottery L_k ($\alpha_k \geq 0$ and $\sum_{k=1}^{K} \alpha_k = 1$)

- What is the probability that you lie on the beach?
- Is there a simple lottery that is similar to the compound lottery (same outcome probabilities)? ("reduced lottery")

Decision making under uncertainty V

Assumption

The decision maker evaluates compound lotteries like their reduced lotteries, i.e. the decision maker is indifferent between a compound lottery and the corresponding reduced lottery.

Decision making under uncertainty VI

axioms preference relation $\succeq$:

- **continuity axiom:** for all lotteries L, L', L'', the sets

$$\{\alpha \in [0,1] : \alpha L + (1-\alpha)L' \succeq L''\}$$

  and

$$\{\alpha \in [0,1] : L'' \succeq \alpha L + (1-\alpha)L'\}$$

  are closed.

- **independence axiom:** for all lotteries L, L', L'' and $\alpha \in (0,1)$ we have

$$L \succeq L' \quad \text{if and only if} \quad \alpha L + (1-\alpha)L'' \succeq \alpha L' + (1-\alpha)L''$$

Decision making under uncertainty VII

Exercise

There are three prices:

1. 2.500.000 \$
2. 500.000 \$
3. 0 \$

An individual prefers the lottery $L_1 = (0.1, 0.8, 0.1)$ to the lottery $L_1' = (0, 1, 0)$.
If the independence axiom is satisfied (as well as transitivity and completeness), can we say which of the following lotteries the individual prefers?
$L_2 = (0.55, 0.4, 0.05)$ $\qquad$ $L_2' = (0.5, 0.5, 0)$

Decision making under uncertainty VIII

Lemma

Assume the independence axiom holds for the preference relation $\succeq$ on the set of lotteries $\mathcal{L}$. Then the following holds:

$$L \sim L' \quad \textit{if and only if} \quad \alpha L + (1-\alpha)L'' \sim \alpha L' + (1-\alpha)L''$$

$$L \succ L' \quad \textit{if and only if} \quad \alpha L + (1-\alpha)L'' \succ \alpha L' + (1-\alpha)L''$$

Proof. (for indifference)

Decision making under uncertainty IX

Lemma

If $L \sim L'$ and $L'' \sim L'''$ and the independence axiom holds, then $\alpha L + (1-\alpha)L'' \sim \alpha L' + (1-\alpha)L'''$ where $\alpha \in [0,1]$.

Proof.

Expected utility theorem I

Definition

A **utility function representing the preferences $\succeq$ on $\mathcal{L}$** is a function $U : \mathcal{L} \to \mathbb{R}$ such that $U(L) \geq U(L')$ whenever $L \succeq L'$ for $L, L' \in \mathcal{L}$.

Definition (von Neumann-Morgenstern utility)

The utility function $U : \mathcal{L} \to \mathbb{R}$ has expected utility form if there is an assignment of numbers $(u_1, \ldots, u_n)$ to the n outcomes in C such that for any simple lottery $(p_1, \ldots, p_n)$ we have

$$U(L) = u_1 p_1 + \cdots + u_n p_n.$$

Such a utility function U is called von Neumann-Morgenstern utility function.

Expected utility theorem II

The idea is that outcome (with certainty) c_i yields utility u_i. To evaluate lotteries, we take the expected utility (i.e. expectation over those u_i).

Expected utility theorem III

Theorem (expected utility theorem)

Assume that the preference relation $\succeq$ satisfies transitivity, completeness, the continuity axiom and the independence axiom. Then $\succeq$ can be represented by a von Neumann-Morgenstern utility function $U : \mathcal{L} \to \mathbb{R}$, i.e. there exists a utility function of the form $U(L) = \sum_{i=1}^{n} u_i p_i$ such that

$$L \succeq L' \quad \text{if and only if } U(L) \geq U(L').$$

Expected utility theorem IV

Proof idea.

- show that there is a worst lottery $\underline{L}$ and a best lottery $\bar{L}$
- assign the worst lottery utility 0 and the best lottery utility 1, i.e. $U(\underline{L}) = 0$ and $U(\bar{L}) = 1$
- L is assigned $U(L) = \alpha$ where α is determined by $L \sim \alpha\bar{L} + (1-\alpha)\underline{L}$

Say, "beach" $\succ$ "stuck at home" $\succ$ "stuck in the hotel room".
Then $u(beach) = 1$ and $u(hotel) = 0$.
$u(\text{home}) = \alpha$ where α is such that

$$\alpha\text{"beach"} + (1-\alpha)\text{"hotel"} \sim \text{"home"}$$

Expected utility theorem V

Proof sketch. (following MWG; not done in detail here)
Step 0: There is a best and worst lottery.

- lotteries putting probability 1 on one outcome
- best/worst of those are better/worse than mixed lotteries (independence axiom!)

Step 1: If $L \succ L'$, then $L \succ \gamma L + (1-\gamma)L' \succ L'$ for any $\gamma \in (0,1)$.
Step 2: Take $\alpha, \beta \in (0,1)$. Then $\beta \bar{L} + (1-\beta)\underline{L} \succ \alpha \bar{L} + (1-\alpha)\underline{L}$ if and only if $\beta > \alpha$.

- trick: $\gamma = (\beta - \alpha)/(1-\alpha)$

Step 3: For every lottery, there is an α such that $[\alpha \bar{L} + (1-\alpha)\underline{L}] \sim L$.

- use closedness of the preferences

Step 4: The utility function $U(L) = \alpha_L$ represents the preference relation.

Expected utility theorem VI

- use step 2

Step 5: The utility function $U(L) = \alpha_L$ is linear, i.e. it is a von Neumann-Morgenstern utility function.

- $$U^{vNM}(L) = p_1\, U(L_1) + p_2\, U(L_2) + \cdots + p_n\, U(L_n)$$

  where I write L_i for the lottery putting all weight on c_i and $L = (p_1, p_2, \ldots, p_n)$ $\square$

Expected utility theorem VII

Exercise

There are three prices:

1. 2.500.000 \$
2. 500.000 \$
3. 0 \$

An individual prefers the lottery $L_1 = (0.1, 0.89, 0.01)$ to the lottery $L_1' = (0, 1, 0)$.
If the individual is an expected utility maximizer (i.e. has a vN-M utility function), can we say which of the following lotteries the individual prefers?
$L_2 = (0, 0.11, 0.89)$ $\quad$ $L_2' = (0.1, 0, 0.9)$

Expected utility theorem VIII

Back to motivation:

- What do these numbers in the game table actually mean?
- What if the other player plays C and D with 50% probability? How to evaluate that?
- can we model a rational decision maker as expected utility maximizer?

	C	D
C	2,2	0,3
D	3,0	1,1

Table: prisoner's dilemma

Review questions

- How is game theory used in economics?
- What do we mean by a "rational, strategic player"?
- Why are we analyzing stylized models instead of real world settings?
- What is a simple lottery? ...a compound lottery?...a reduced lottery?
- State and interpret the continuity and independence axiom!
- What is a von Neumann - Morgenstern utility function? Under which assumptions/axioms can preferences be represented by a von Neumann - Morgenstern utility function?
- Why are von Neumann - Morgenstern utility functions important for game theory?

reading: MWG p. 167–178 or (alternatively) MSZ p. 9–23
* reading: Gilboa, Itzhak, et al. "Economic models as analogies." (link on website), MWG p. 178–182

Exercises I

1. L_1 be throwing a fair dice: $C = \{1, 2, 3, 4, 5, 6\}$, $p_i = 1/6$ for $i = 1, \ldots, 6$
   L_2 be throwing an "unfair dice": $p_1 = p_2 = 1/4$ and $p_3 = p_4 = p_5 = p_6 = 1/8$
   What is the probability that the outcome "1" occurs in the compound lottery $(L_1, L_2; 1/2, 1/2)$? Derive the reduced lottery of the compound lottery $(L_1, L_2; 1/2, 1/2)$.
2. There are three prices:
   1. 2.500.000 \$
   2. 500.000 \$
   3. 0 \$

Exercises II

An individual prefers the lottery $L_1 = (0.1, 0.8, 0.1)$ to the lottery $L_1' = (0, 1, 0)$.
If the independence axiom is satisfied (as well as transitivity and completeness), which of the following lotteries does the individual prefer?
$L_2 = (0.35, 0.6, 0.05) \qquad L_2' = (0.3, 0.7, 0)$

3. Prove the strict preference part of the lemma on slide 19.